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VEHICLE EXTERIOR CLOSURE SOUND AMPLIFICATION AND DIRECTIONAL SPEAKER SYSTEM

Abstract

Directional audio is provided to positions around the vehicle. A position mapping is stored defining information indicative of how each of a plurality of closures of a vehicle, in a plurality of positions, affects sound emanating from a plurality of speakers of the vehicle at different positions around the exterior of the vehicle. Respective locations of at least one user outside the vehicle are tracked. The position mapping is utilized to apply power levels to the plurality of speakers and positioning to the plurality of the closures to direct sound output from the speakers to the respective locations of the at least one user at a desired volume level.

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Background/Summary

TECHNICAL FIELD

[0001] Aspects of the disclosure generally relate to a vehicle exterior closure sound amplification and directional speaker system.

BACKGROUND

[0002] Vehicles may have speakers inside and outside the vehicle cabin. Vehicles may also have different numbers and types of closures, such as doors, windows, sunroofs, etc. Some vehicles may also have passive entry features that allow the vehicle to detect a key fob or phone in proximity to the vehicle. Further, some vehicles may also have exterior cameras that allow the vehicle to detect and track a person in proximity to the vehicle.

SUMMARY

[0003] In one or more illustrative embodiments, a method for providing audio to positions around the vehicle includes: storing a position mapping defining information indicative of how each of a plurality of closures of a vehicle, in a plurality of positions, affects sound emanating from a plurality of speakers of the vehicle at different positions around the exterior of the vehicle; tracking respective locations of at least one user outside the vehicle; and utilizing the position mapping to apply power levels to the plurality of speakers and positioning to the plurality of the closures to direct sound output from the speakers to the respective locations of the at least one user at a desired volume level.

[0004] In one or more illustrative embodiments, a vehicle for use in providing audio to positions around the vehicle includes one or more controllers configured to: store a position mapping defining information indicative of how each of a plurality of closures of the vehicle, in a plurality of positions, affects sound emanating from a plurality of speakers of the vehicle at different positions around the exterior of the vehicle, the plurality of positions including a plurality of intermediate positions between an open position and a closed position, track respective locations of at least one user outside the vehicle, and utilize the position mapping to apply power levels to the plurality of speakers and positioning to the plurality of the closures to direct sound output from the speakers to the respective locations of the at least one user at a desired volume level.

[0005] In one or more illustrative embodiments, a non-transitory computer-readable medium comprising instructions for providing audio to positions around a vehicle that, when executed by one or more controllers of the vehicle, cause the vehicle to perform operations including to store a position mapping defining information indicative of how each of a plurality of closures of the vehicle, in a plurality of positions, affects sound emanating from a plurality of speakers of the vehicle at different positions around the exterior of the vehicle, the plurality of positions including a plurality of intermediate positions between an open position and a closed position, track respective locations of at least one user outside the vehicle, and utilize the position mapping to apply power levels to the plurality of speakers and positioning to the plurality of the closures to direct sound output from the speakers to the respective locations of the at least one user at a desired volume level.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** illustrates an example vehicle for use in a system providing directional audio features to positions around the vehicle;

[0007] FIG. **2** illustrates an example testing setup for characterizing a vehicle to create the position mapping;

[0008] FIG. **3** illustrates an example of operation of the system for providing directional audio features to a plurality of tracking devices at positions around the vehicle;
[0009] FIG. **4** illustrates an example process for characterizing a vehicle to create the position mapping;
[0010] FIG. **5** illustrates an example process for providing directional audio features to tracking devices at positions around the vehicle; and
[0011] FIG. **6** illustrates an example of a computing device for use in providing directional audio features to positions around the vehicle.

DETAILED DESCRIPTION

[0012] As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

[0013] Vehicles may have built-in speakers and as well as outlets to power add-on speakers. Surrounding many of these speakers and outlets are exterior closures such as doors, windows, frunk, liftgate, tailgate, and trunk lids. These closures may be controllable by the vehicle and/or may be positionable in open, closed, and/or any set position in-between. The positioning of the closures may be controlled to allow the sound that is produced by the speakers to be reflected towards specific positions outside of the vehicle. This reflection of sound may be used to amplify and focus the sound produced by the speakers. In some examples, the sensor suite of the vehicle may be used to identify the relative position of listeners with respect to the vehicle. The sound from the speakers may be aimed using the closures to optimize the listening experience of the listeners at those relative positions. Further aspects of the disclosure are discussed in detail herein.

[0014] FIG. **1** illustrates an example vehicle **102** for use in a system **100** providing directional audio features to positions around the vehicle **102**. The vehicle **102** may include various speakers **106** configured to convert electrical signals into audible sounds. These speakers **106** may be located within the cabin, trunk, frunk, truck bed, and other compartments of the vehicle **102**. The vehicle **102** may also include various closures **108**, such as doors, windows, lids, etc. configured to be controllably opened and closed to provide access to the various compartments. The vehicle **102** may also include one or more controllers configured to utilize sensors **128** to track the users. Based on the positions of the users, the one or more controllers may be configured to provide audio signals to the speakers **106** and manipulate the closures **108** to selectively direct audio in relation to the users.

[0015] As shown, the vehicle **102** is a sport utility vehicle (SUV). However, the vehicle **102** may include any of various types of automobile, crossover utility vehicle (CUV), SUV, truck, recreational vehicle, boat, plane or other mobile machine for transporting people or goods. Such vehicles **102** may be human-driven or autonomous. In many cases, the vehicle **102** may be powered by an internal combustion engine. As another possibility, the vehicle **102** may be a battery electric vehicle powered by one or more electric motors. As a further possibility, the vehicle **102** may be a hybrid electric vehicle powered by both an internal combustion engine and one or more electric motors, such as a series hybrid electric vehicle, a parallel hybrid electrical vehicle, or a parallel/series hybrid electric vehicle.

[0016] The vehicle **102** may include one or more controllers configured to perform and manage various vehicle **102** functions under the power of the vehicle battery and/or drivetrain. The controllers may include various types of computing devices in support of performance of the functions of the controllers described herein. In an example, the controllers may include one or more processors configured to execute computer instructions, and a storage medium on which the computer-executable instructions and/or data may be maintained. A computer-readable storage medium (also referred to as a processor-readable medium or storage) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by the processor(s)). In general, the processor receives instructions and/or data, e.g., from the storage, etc., to a memory and executes the instructions using the data, thereby performing one or more processes, including one or more of the processes described herein. Computer-executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java, C, C++, C#, Fortran, Pascal, Visual Basic, Python, JavaScript, Perl, etc.

[0017] As depicted, the example vehicle controllers are represented as discrete controllers (e.g., an audio system **104**, a powertrain controller **110**, a body controller **112**, a location controller **120**, a human machine interface (HMI) controller **122**, and a telematics controller **124**). However, the vehicle controllers may share physical hardware, firmware, and/or software, such that the functionality from multiple controllers may be integrated into a single controller, and that the functionality of various such controllers may be distributed across a plurality of controllers.

[0018] The audio system **104** may be configured to control various speakers **106** audible outside of the exterior of the vehicle **102**. These speakers **106** may include speakers **106** that are built-into the vehicle **102**, as well as speakers **106** that are add-ons that are added to the vehicle **102**. As some examples, the speakers **106** may include speakers **106** within the doors of the vehicle **102**, speakers **106** within the headrests or headliner of the vehicle **102**, speakers **106** within the front and/or rear dash of the vehicle **102**, speakers **106** within the frunk of the vehicle **102**, speakers **106** within the trunk or bed of the vehicle **102**, etc.

[0019] The vehicle **102** may define various closures **108** that allow for access to various compartments of the vehicle **102** in which speakers **106** are located. These closures **108** may include, as some non-limiting examples, doors to the cabin, a tailgate to the truck bed (e.g., of a truck bed covered by a tonneau cover), hoods to the trunk and/or frunk, a sunroof, a decklid, a liftgate, etc.

[0020] The powertrain controller **110** may be configured to provide control of engine operating components (e.g., idle control components, fuel delivery components, emissions control components, etc.) and for monitoring status of such engine operating components (e.g., status of engine codes).

[0021] The body controller **112** may be configured to manage the locking, unlocking of various exterior closures **108** of the vehicle **102**, keyless entry, remote start, and point of access status verification (e.g., closure **108** status of the hood, doors and/or trunk of the vehicle **102**). In many examples, the closures **108** may be motorized or otherwise automated. In such examples, the body controller **112** may be configured to control and/or monitor the open or closed status of the closures **108**. For example, the body controller **112** may be configured to control window actuators to open or close windows of the vehicle **102**. In another example, the body controller **112** may control trunk or frunk actuators to open or close the trunk lid and/or frunk lid of the vehicle **102**. In yet another example, the vehicle **102** may have actuators for automatically opening and closing doors, and the body controller **112** may control the angle of the opening or closing of the doors.

[0022] The body controller **112** may also be configured to manage various other functions such as exterior lighting **114**. The exterior lighting **114** may include various lights on the exterior of the vehicle **102**. These may include, for example, headlights that shine forwards in front of the vehicle **102** (e.g., low and/or high beams), taillights that shine rearwards behind the vehicle **102**, a center high mounted stop light (CHMSL), curtain lighting that shines to the sides of the vehicle **102**, bed lights in the bed of a truck bed, and/or puddle lamps that shine downwards from under the side mirrors. Some vehicles **102** may have aimable headlights that can be steered into different directions. Also, some vehicles **102** may be equipped with dynamic light emitting diode (LED) headlights that can shape the light output. Some vehicles **102** (e.g., police cruisers or off-road vehicles **102**) may also have spotlights or other aftermarket lights that may be aimable or otherwise controllable. The exterior lighting **114** of the vehicle **102** may be adjusted in orientation and intensity by the body controller **112**.

[0023] The body controller **112** may be in communication with one or more wireless transceivers **116**. The wireless transceiver **116** may be configured to facilitate communication with tracking devices **118** using wireless protocols such as Bluetooth Low Energy (BLE) or ultra-wideband (UWB). The tracking devices **118** may include key fobs, mobile phones, or dedicated devices that are configured to communicate wirelessly with the wireless transceiver **116** to facilitate identification and positioning of a user. The wireless transceiver **116** may allow the body controller **112** to identify the positions of key fobs, mobile phones, or other devices that may identify users for access to the vehicle **102**. In an example, the body controller **112** may unlock doors of the vehicle **102** responsive to detection of an approach of an authorized user via the wireless transceiver **116** (e.g., detected based on increased BLE signal strength or decreasing Radio Frequency Time-of-Flight (RF-ToF) of the approaching tracking device **118**).

[0024] The location controller **120** may be configured to provide vehicle location information. For example, the location controller **120** may allow the vehicle **102** to receive time and location information from a global navigation satellite system (GNSS) constellation of satellites.

[0025] The HMI controller **122** may be configured to receive user input via various buttons or other controls, as well as provide vehicle status information to a driver, such as fuel or charge level information, engine operating temperature information, and current location of the vehicle **102**. The HMI controller **122** may be configured to provide information to various displays within the vehicle **102**, such as a center stack touchscreen, a gauge cluster screen, etc.

[0026] The telematics controller **124**, sometimes referred to as a telematics control unit (TCU), may include network hardware configured to facilitate communication between the other vehicle controllers and with other devices of the system **100**. The telematics controller **124** may include or otherwise access a modem **126** configured to facilitate communication with other vehicles **102** or with infrastructure. The modem **126** may, additionally, be configured to communicate over a broadcast protocol to facilitate cellular vehicle-to-everything (C-V2X) communications with devices such as other vehicles **102**. The telematics controller **124** may be further configured to communicate over various other protocols, such as with a communication network over a network protocol. It should be noted that these protocols are merely examples, and different peer-to-peer and/or cellular technologies may be used.

[0027] The vehicle **102** may also make use of various sensors **128** in order to receive information with respect to the surroundings of the vehicle **102**. In an example, these sensors **128** may include one or more of cameras (e.g., advanced driver assistance system (ADAS) cameras), ultrasonic sensors, radar systems, and/or lidar systems. The sensors **128** may be used to allow the vehicle **102** to image its surroundings. For instance, camera sensors **128** mounted on the front, rear, and sides of the vehicle **102** may be used to capture visual images of the surroundings of the vehicle **102**. These images may be used, for example, to detect the positions of users who are not carrying tracking device **118**, to confirm the positions of users who are carrying tracking device **118**, to identify the positions of objects surrounding the vehicle **102**, etc.

[0028] A vehicle bus **130** may include various methods of communication available between the various components of the vehicle **102** discussed herein. As some non-limiting examples, the vehicle bus **130** may include one or more of a vehicle controller area network (CAN), an Ethernet network, and a media-oriented system transfer (MOST) network. The vehicle bus **130** may also be wireless. While a single vehicle bus **130** is illustrated, it should be noted that in many examples, multiple vehicle buses **130** are included, with a subset of the controllers connected to each vehicle bus **130**.

[0029] The positions of users around the vehicle **102** may be tracked using the tracking devices **118** and/or using the sensors **128** of the vehicle **102**. Based on these positions, the speakers **106** may be activated and the closures **108** of the vehicle **102** may be opened or closed or positioned to direct sound to the detected positions of the users.

[0030] To determine the correct settings for the speakers **106** and the closures **108**, the vehicle **102** may maintain a position mapping **132**. The position mapping **132** may include information indicative of how each closure **108**, in a variety of positions, will affect the sound emanating from the speakers **106**, such as in terms of loudness, clarity, pitch, resolution, or other related sound quality metrics for potential listeners around the vehicle **102** at varying distances. Given the position of the user, the position mapping **132** may accessed to determine the correct closure **108** and speaker **106** settings to direct the sound to the detected user positions. Additionally, or alternately, the vehicle **102** itself may be rotated, oriented, and/or moved to direct the sound to the detected user positions.

[0031] FIG. **2** illustrates an example testing setup **200** for characterizing a vehicle **102** to create the position mapping **132**. The vehicle **102** includes a plurality of speakers **106** and a plurality of closures **108**. As shown in the example vehicle **102**, a first speaker **106**A is located in a frunk with a first closure **108**A, a second speaker **106**B is located in a driver side door closure **108**B, a third speaker **106**C is located in a passenger side door closure **108**C, a fourth speaker **106**D is located in a driver side second row door closure **108**D, a fifth speaker **106**E is located in a passenger side second row door closure **108**E, and a sixth speaker **106**F is located in a rear cargo area in proximity to a rear liftgate closure **108**F. A sunroof closure **108**G is also provided in the cab of the vehicle **102**. It should be noted that this is only one example of a vehicle **102**, and vehicles **102** having more, fewer, and/or differently located speakers **106** and closures **108** are possible. For instance, if the vehicle **102** is a standard cab pickup truck or a coupe, then the vehicle **102** may have only two doors. Or if, the vehicle **102** is a minivan, then the vehicle **102** may have one or more sliding side doors.

[0032] One or more testing locations **202** may be used to characterize the sound output of the speakers **106** for various open and closed positions of the closure **108**. As shown testing locations **202**A-L are placed in an ellipse around a vehicle **102** at a given distance. Signals of a predefined power may be applied to the speakers **106**, and testing devices may be used to measure the sound pressure at the different positions of the testing locations **202** (the example testing locations **202** for measurement being represented as microphones for sake of illustration). In an example, the tracking devices **118** themselves may be used as the testing devices. In another example, one or more other mobile devices may be used as the testing devices. In yet another example, one or more microphones or other sound capture devices available at a factory or audio laboratory may be used as the testing devices. It should also be noted that while a plurality of testing locations **202** are shown, a single testing device or multiple testing devise may be used and/or moved to the various different testing locations **202** to perform the testing.

[0033] The captured information may be used to characterize the sound output of the vehicle **102**. For instance, given one or more predefined power levels to each speaker **106** with the closures **108** closed, the frequency response of the vehicle **102** may be captured at the various testing locations **202**. Then, this measurement may be repeated per speaker **106**, after changing the positions of the closures **108**.

[0034] For instance, a sequence of measurements may be taken of each speaker **106**B, **106**C separately and/or in combination with the driver side door closure **108**B and the passenger side door closure **108**C in closed positions and the sunroof closure **108**G at a plurality of different positions (shown in the FIG. **2** as dotted lines). Similar, a sequence of measurements may be taken of each speaker **106**B, **106**C separately and/or in combination with the sunroof closure **108**G closed and one of the driver side door closure **108**B, the passenger side door closure **108**C, the driver side second row door closure **108**D, the passenger side second row door closure **108**E, the rear liftgate closure **108**F, etc., at a plurality of different positions. This process may be repeated, as desired, for any and all different combination of speakers **106** and/or closure **108** positions (e.g., opening the sunroof, opening multiple doors, opening the liftgate and a single door). The process may also be repeated after changing the distance between the testing locations **202** and the vehicle **102**, to provide information with respect to distance as well.

[0035] As the position mapping **132** indicates the sound output at different positions around the vehicle **102** given different combinations of speaker **106** power level and closure **108** position, the position mapping **132** may be used to look up a specific power level at a position around the vehicle **102** to retrieve the speaker **106** power level and closure **108** position settings to use to achieve that result.

[0036] In some examples, the information in the position mapping **132** may be extrapolated based on identities such as that doubling of distance reduces the sound level by 6 dB (e.g., to reduce the quantity of distances from the vehicle **102** that are measured). Thus, settings may also be inferred based on the position mapping **132**, despite those specific settings not having been recorded to the position mapping **132**. In another example, a machine learning model may be trained based on the combinations of speaker **106** power level and closure **108** position closure **108** position and ground truth recorded sound level to infer the correct settings for speaker **106** power level and closure **108** position to use in various situations.

[0037] In some examples, the speakers **106** may include add-on speakers **106** connected to power outlets of the vehicle **102**. In such a case, performance metrics for the add-on speakers **106** may be communicated to the vehicle **102** (e.g., via the wireless transceiver **116**) to allow the vehicle **102** to control the loudness and other parameters of the add-on speakers **106** more accurately. The user may also designate which built-in and/or add-on speakers **106** are to be active to limit energy consumption. In the case where add-on speakers **106** are to be used, the locations of the add-on speakers **106** may also be indicated to the vehicle **102**, e.g., based on which power outlet the add-on speakers **106** are connected to, based on user input of the orientation of the add-on speakers **106**, etc.

[0038] FIG. **3** illustrates an example **300** of operation of the system **100** for providing directional audio features to a plurality of tracking devices **118** at positions around the vehicle **102**. As shown, a first tracking device **118**A (e.g., a fob) is located in front of the vehicle **102**, a second tracking device **118**B (e.g., a phone) is located on the driver side of the vehicle **102**, and a third tracking device **118**C (e.g., another phone) is located on the passenger side of the vehicle **102**.

[0039] In an example, the locations of the tracking devices **118**A, **118**B, **118**C may be identified by the vehicle **102** using the wireless transceiver **116**. This may be done using various techniques, such as wireless trilateration, ultrasonic audio trilateration, receipt of GNSS coordinates from the tracking device **118** (at least for a coarse positioning depending on available GNSS precision), image recognition of the users using the sensors **128** of the vehicle **102**, etc.

[0040] Sound may be provided to the users holding the tracking devices **118** based on various settings of the vehicle **102**. In an example, the owner of the vehicle **102** may request for the vehicle **102** to play a song, play audio of a sports game, play streamed media content, play a streamed radio station, play a broadcast radio station, etc. This may be done, for example, using the HMI of the vehicle **102**, via a request to the vehicle **102** from the user's smartphone (e.g., also the user's tracking device **118**), etc. The user may also set a volume to be used to reproduce the audio. This may again be set via the HMI of the vehicle **102** (such as a volume knob), or via the smartphone. Once set, the user may wander around the vehicle **102**, and the vehicle **102** may track the user and direct the sound to the position of the user.

[0041] As shown, the vehicle **102** is tracking three different tracking devices **118**A, **118**B, **118**C. In one example, the vehicle **102** may use the closest speaker **106** (or speakers **106**) to each tracking device **118** to provide audio to that position. For instance, the vehicle **102** may use the speaker **106**A and adjust the position of the closure **108**A to provide audio to the position of the tracking device **118**A, the vehicle **102** may use the speaker **106**B and adjust the position of the closure **108**B to provide audio to the position of the tracking device **118**B, and the vehicle **102** may use the speaker **106**C and adjust the position of the closure **108**C to provide audio to the position of the tracking device **118**C.

[0042] It should be noted that, as each of the different tracking devices **118**A, **118**B, **118**C is receiving audio from a different speaker **106**, different audio content may be provided to each of the users. For instance, the speaker **106**A may provide first audio content to a first sound zone **302**A encompassing the position of the tracking device **118**A, the speaker **106**A may provide second audio content to a second sound zone **302**B encompassing the position of the tracking device **118**B, and the speaker **106**C may provide third audio content to a third sound zone **302**C encompassing the position of the tracking device **118**C.

[0043] In another example, the vehicle **102** may provide audio content to one of the users, while using the remaining speakers **106** to provide white noise to mask the audio content, thereby giving privacy from other potential listeners to the user intended to hear the audio content. As one example, the user of tracking device **118**B may wish to use the vehicle **102** to receive audio from a confidential meeting or audio file. In such a case, the speaker **106**B and closure **108**B may be used to direct the media content to the position of the tracking device **118**B, while the other speakers **106**A, **106**C, **106**D may be used to provide white noise surrounding the vehicle **102** to mask the media content provided to the position of the tracking device **118**B. For instance, the speaker **106**A may be targeted to provide white noise to the position of the tracking device **118**A in the first sound zone **302**A, the speaker **106**C may be targeted to provide white noise to the position of the tracking device **118**C in the third sound zone **302**C, and the speaker **106**F may be generally set to provide white noise to the surroundings in a fourth sound zone **302**D without a specific identified target.

[0044] FIG. **4** illustrates an example process **400** for characterizing a vehicle **102** to create the position mapping **132**. In an example, the process **400** may be performed by the one or more controllers of the vehicle **102**, in the context of the system **100**.

[0045] At operation **402**, the vehicle **102** captures measurements of the speakers **106** of the vehicle **102**. These measurements are taken with at different testing locations **202** around the exterior of the vehicle **102**, e.g., using various testing devices such as the tracking devices **118**, microphones, etc. Signals of a predefined power may be applied to the speakers **106**, and the testing devices may be used to measure the sound pressure at the different testing locations **202**. The captured information may be used to characterize the sound output of the vehicle **102**. For instance, given one or more predefined power levels to each speaker **106** with the closures **108** closed, the frequency response of the vehicle **102** may be recorded. Then, this measurement may be repeated per speaker **106**, after changing the positions of the closures **108**.

[0046] At operation **404**, the vehicle **102** constructs a position mapping **132**. Based on the captured measurements, the position mapping **132** may include a mapping of sound output at different positions around the vehicle **102** given different combinations of speaker **106** power level and closure **108** position. Thus, the position mapping **132** may be usable to allow the vehicle **102** to look up settings for the speakers **106** and closure **108** positions to direct sound to a location of a user at a predefined loudness. In some examples, the information in the position mapping **132** may be extrapolated based on identities such as that doubling of distance reduces the sound level by 6 dB. Thus, settings may also be inferred based on the position mapping **132**, despite those specific settings not having been recorded to the position mapping **132**. In another example, a machine learning model may be trained based on the combinations of speaker **106** power level and closure **108** position and ground truth recorded sound level to infer the correct settings for speaker **106** power level and closure **108** position to use in various situations.

[0047] In yet another example, for add-on speakers **106** the vehicle **102** may receive a characterization of the properties of the add-on speaker **106** wirelessly from the add-on speaker **106**. Or, the vehicle **102** may receive an identifier of a model of the add-on speaker **106**, and may download or otherwise access characterization of the properties of the add-on speaker **106** based on the identifier.

[0048] At operation **406**, the position mapping **132** is stored to the vehicle **102**. In an example, the position mapping **132** may be stored to the body controller **112**. In another example, the position mapping **132** may be stored to another controller, such as to the audio system **104** itself. After operation **406**, the process **400** ends.

[0049] Variations on the process **400** are possible. In another example, the capture portion of the process **400** may be performed by a computing device external to the vehicle **102**, and the results may be loaded to the vehicle **102** later for use. This loading may be performed during build of the vehicle **102**, via an over-the-air update, etc.

[0050] FIG. **5** illustrates an example process **500** for providing directional audio features to tracking devices **118** at positions around the vehicle **102**. In an example, the process **500** may be performed by the one or more controllers of the vehicle **102**, in the context of the system **100**.

[0051] At operation **502**, the vehicle **102** sets up media content for playback by the vehicle **102**. In an example, the owner of the vehicle **102** may request for the vehicle **102** to play a song, play audio of a sports game, play streamed media content, play a streamed radio station, play a broadcast radio station, etc. This may be done, for example, using the HMI of the vehicle **102**, via a request to the vehicle **102** from the user's smartphone (e.g., also the user's tracking device **118**), etc. The user may also set a volume to be used to reproduce the audio. This may again be set via the HMI of the vehicle **102** (such as a volume knob), or via the smartphone. Once set, the user may wander around the vehicle **102**, and the vehicle **102** may track the user and direct the sound to the position of the user.

[0052] At operation **504**, the vehicle **102** tracks positions of the users. In an example, the locations of the tracking devices **118** may be identified by the vehicle **102** using the wireless transceiver **116**. This may be done using various techniques, such as trilateration, receipt of GNSS coordinates from the tracking device **118**, image recognition of the users using the sensors **128** of the vehicle **102**, etc.

[0053] At operation **506**, the vehicle **102** utilizes the position mapping **132** to identify speaker **106** and closure **108** settings for the vehicle **102** to play back the media content to the identified positions of the tracking devices **118**.

[0054] At operation **508**, the vehicle **102** applies the settings to the speakers **106** and closures **108** of the vehicle **102**. Accordingly, the vehicle **102** applies power levels to the plurality of speakers **106** and positioning to the plurality of the closures **108** to direct sound output from the speakers **106** to the position of the user. After operation **508**, the process **500** ends.

[0055] Variations on the process **500** are possible. In some examples, the vehicle **102** can use air suspension to tilt the vehicle **102** to improve speaker **106** orientation to the locations of the tracking devices **118**. In another example, the vehicle **102** can utilize semiautonomous driving and/or a tank turn steering function to reposition and/or rotate the vehicle **102** such that the more powerful speakers **106** are facing the listeners, e.g., if there is a loud volume audio request. In yet a further example, the vehicle **102** may maneuver or otherwise be repositioned either fully autonomously, semi-autonomously, via suggested remote-control action performed using commands from the user's tracking device **118**, and/or by providing a visual indication and/or instruction of the desired location to a display of the vehicle **102** on the user's tracking device **118**. These actions may be performed at operation **506** in combination with, or instead of movement of the closures **108**.

[0056] In another variation, one or more closures **108** may be manually controlled. In such a variation, those closures **108** may not be manipulable by the vehicle **102**. In such examples, the position mapping **132** may still be defined to account for these variations. Then, the vehicle **102** can choose the sound levels to apply to the speakers **106** and in which direction to open the automated closures **108** (if any), based on the position mapping **132**.

[0057] In another variation, one or more closures **108** may be hinged in multiple ways. In such a variation, the closures **108** may be manipulable by the vehicle **102** and/or otherwise positionable in multiple directions. For instance, a station wagon style tailgate may be employed that is side hinged to open as a door and also bottom hinged to open as a tailgate. In another example, a vehicle door may be hinged on either side to open either on the left side or the right side. In yet another example a door may be hinged to open downwards as a tailgate or, in the alternative, in the style of two barn doors hinged at either side opposite. In such examples, the position mapping **132** may be defined to account for these variations. Then, the vehicle **102** can choose in which direction to open the closures **108**, in addition to choosing which closures **108** to use, how much to open the closures **108**, and the sound levels to apply to the speakers **106**.

[0058] In an example, the vehicle **102** may be configured to allow sound connection from third parties. In one non-limiting example, a police or other priority user may be able to message the vehicle **102**, e.g., via the wireless transceiver **116**, to commandeer the audio system **104** of the vehicle **102** for various purposes. The vehicle **102** may also close certain closures **108** and/or warn users if unauthorized individuals are approaching the vehicle **102** and/or a storm is incoming.

[0059] In another example, the exterior lighting **114** of the vehicle **102** may be utilized in addition to the directional audio features. In an example, the exterior lighting **114** may be synchronized with the media content being played back to music to provide additional ambiance. For instance, the exterior lighting **114** may be configured to operate as level meters, such that the lighting intensity and/or color corresponds to the relative loudness of the media content being played back. In such an example, the lighting intensity is raised for louder sounds and reduced for quieter sounds (or the reverse) and/or shifted in color (e.g., lower frequencies of light such as red for loudest and higher frequencies of light such as green or blue for quietest, or some other combination of color change). In another example, the exterior lighting **114** may be aimed towards the tracking devices **118** to allow the locations of the users listening to the media content to be observed. In some cases, this feature may be available provided that the vehicle **102** has at least a minimum threshold state of charge, and may be disabled otherwise, to avoid running the battery of the vehicle **102** down due to the power requirements inherent in lighting.

[0060] In yet another example, key-off load optimization may be provided in combination with the directional audio features. For instance, depending on which speakers **106** are being leveraged, audio media can be converted from a higher channel representation to a lower channel representation. For instance, surround sound audio may be converted to two-channel stereo or mono to enable only the most effective speakers **106** contribute to providing sound, without a loss of audio quality such as tracks that rely on stereo to introduce different instruments/sounds to the right vs. left.

[0061] In another example of key-off load optimization, traditional lighting operating patterns of the vehicle **102** may be altered to reduce electrical load during usage of the directional audio features. For instance, exterior and cabin courtesy lighting may be disabled as is the default during daylight hours. During post-sunset conditions, an overall lighting scheme may be determined by the vehicle **102** through a load arbitration process to allow some light for users while minimizing the load on the electrical system.

[0062] In some cases, a feature usage duration may be communicated to the user via the HMI of the vehicle **102** or via an application installed to the user's tracking device **118**. This user interface may allow the user to configure their usage based on the duration of intended use of the directional audio features. For instance, the user interface may indicate, based on a measurement of battery state of charge and an estimate of the power draw of the desired lighting and audio features, one or more of: a computed number of minutes that full volume capability plus music synchronized lighting may be supportable given the current battery level, a computed number of minutes that full volume capability without lighting may be supportable given the current battery level, a computed number of minutes that a set volume capability with or without plus music synchronized lighting may be supportable given the current battery level, etc. In another possibility, the user interface may support an auto mode which allows user to specify a desired duration for operation of the directed sound feature and the vehicle **102** may start with full functionality for specific time and taper off features over time to meet the desired duration without dropping below a minimum threshold state of charge.

[0063] The vehicle **102** may also use one or more additional load optimization features. For example, the vehicle **102** may disable the heating, ventilation, and air conditioning (HVAC) system while the directional audio features are being used. In another example, the vehicle **102** may disable all HMI chimes while the directional audio features are being used. In another example, the vehicle **102** may set all in-vehicle HMI screens and lighting to a lowest brightness level or to off (e.g., with touch-wake up) while the directional audio features are being used. In another example, the vehicle **102** may slow down sampling of door ajar circuits to conserve power while the directional audio features are being used. In another example, the vehicle **102** may dim or shut off the cabin courtesy lights and/or shut off the cabin lights except for the courtesy lights on or near the open door while the directional audio features are being used.

[0064] In yet another example of a variation of use of the directional audio features, the vehicle **102** may rebalance the audio to speakers **106** to help best project the stereo soundstage. For instance, if the sound is being projected from an open driver door, the driver door speaker **106** may be set to deliver sound at a sound pressure level (SPL) below that of the far passenger door that is closed. Otherwise, a stereo effect from the content may be lost.

[0065] In another example, the speakers **106** may include body panel speakers **106** which can be employed where an exciter is mounted to the back surface of the body panel and used to communicate sound waves via the body panels themselves. In such an example, the body panels in the direction of the interested audience will be energized to project sound in that direction.

[0066] In another example, gesture recognition may be implemented by the vehicle **102** implementing the directional sound features to offer a touch-free user interface. This interface may be available to users of the tracking devices **118**. In another example, the interface may be open and offered to people who are passing by or otherwise in the area of the vehicle **102** who either do or do not want to be exposed to the sound. For example, the gestures that are available for controlling the vehicle **102** may include, for example, a first gesture to increase the volume (e.g., raising the user's hand), a second gesture to decrease the volume (e.g., lowering the user's hand), a third gesture to reduce a white noise source (e.g., putting one's fingers in one's ear). This gesture interface may be implemented using data captured by the exterior sensors **128** of the vehicle **102**. For instance, the external sensors **128** may capture image data, and image recognition may be performed by the vehicle **102** based on the captured image data to identify the gestures performed by the user. In another example, UWB external sensors **128** may capture wireless device location data (e.g., of a smartwatch or other device of the user, or even of the user directly), and the gesture recognition may be performed by the vehicle **102** based on tracking the location of the device and/or user via UWB.

[0067] In another example, vehicles **102** implementing the directional sound features may be used as a paired speaker **106**. In such an approach, surrounding vehicles **102** may be employed as additional speakers **106** through an application installed to the user's tracking device **118**, where people can employ a parked vehicle **102** to act as a speaker **106**, e.g., in exchange for tokens or other compensation. The vehicles **102** may also communicate their position and the position of listeners to one another to allow the vehicles **102** to collectively implement a surround sound experience. For instance, the vehicles **102** on the left side of a parking lot may join together to provide the left channel output while the vehicles **102** on the right side of a parking lot may join together to provide the right channel output. Additionally, using the positions of the vehicles **102**, the vehicles **102** may synchronize their outputs for a target user location, e.g., as determined via the tracking device **118**, to be synched temporally with the speed of sound through air up based on their distances from the user to create a more immersive experience.

[0068] FIG. **6** illustrates an example **600** of a computing device **602** for use in providing directional audio features to positions around the vehicle **102**. Referring to FIG. **6**, and with reference to FIGS. **1**-**5**, the controllers of the vehicles **102** and the tracking devices **118** may be examples including such computing devices **602**. As shown, the computing device **602** includes a processor **604** that is operatively connected to a storage **606**, a network device **608**, an output device **610**, and an input device **612**. It should be noted that this is merely an example, and computing devices **602** with more, fewer, or different components may be used.

[0069] The processor **604** may include one or more integrated circuits that implement the functionality of a central processing unit (CPU) and/or graphics processing unit (GPU). In some examples, the processors **604** are a system on a chip (SoC) that integrates the functionality of the CPU and GPU. The SoC may optionally include other components such as, for example, the storage **606** and the network device **608** into a single integrated device. In other examples, the CPU and GPU are connected to each other via a peripheral connection device such as peripheral component interconnect (PCI) express or another suitable peripheral data connection. In one example, the CPU is a commercially available central processing device that implements an instruction set such as one of the x86, ARM, Power, or microprocessor without interlocked pipeline stage (MIPS) instruction set families.

[0070] Regardless of the specifics, during operation the processor **604** executes stored program instructions that are retrieved from the storage **606**. The stored program instructions, accordingly, include software that controls the operation of the processors **604** to perform the operations described herein. The storage **606** may include both non-volatile memory and volatile memory devices. The non-volatile memory includes solid-state memories, such as not and (NAND) flash memory, magnetic and optical storage media, or any other suitable data storage device that retains data when the system is deactivated or loses electrical power. The volatile memory includes static and dynamic random-access memory (RAM) that stores program instructions and data during operation of the system **100**.

[0071] The GPU may include hardware and software for display of at least two-dimensional (2D) and optionally three-dimensional (3D) graphics to the output device **610**. The output device **610** may include a graphical or visual display device, such as an electronic display screen, projector, printer, or any other suitable device that reproduces a graphical display. As another example, the output device **610** may include an audio device, such as a loudspeaker or headphone. As yet a further example, the output device **610** may include a tactile device, such as a mechanically raiseable device that may, in an example, be configured to display braille or another physical output that may be touched to provide information to a user.

[0072] The input device **612** may include any of various devices that enable the computing device **602** to receive control input from users. Examples of suitable input devices **612** that receive human interface inputs may include keyboards, mice, trackballs, touchscreens, microphones, graphics tablets, and the like.

[0073] The network devices **608** may each include any of various devices that enable the vehicles **102** and tracking devices **118** to send and/or receive data from external devices over networks. Examples of suitable network devices **608** include an Ethernet interface, a Wi-Fi transceiver, a cellular transceiver, or a BLUETOOTH or BLE transceiver, an UWB transceiver or other network adapter or peripheral interconnection device that receives data from another computer or external data storage device, which can be useful for receiving large sets of data in an efficient manner.

[0074] The processes, methods, or algorithms disclosed herein can be deliverable to/implemented by a processing device, controller, or computer, which can include any existing programmable electronic control unit or dedicated electronic control unit. Similarly, the processes, methods, or algorithms can be stored as data and instructions executable by a controller or computer in many forms including, but not limited to, information permanently stored on non-writable storage media such as read-only memory (ROM) devices and information alterably stored on writeable storage media such as floppy disks, magnetic tapes, compact discs (CDs), RAM devices, and other magnetic and optical media. The processes, methods, or algorithms can also be implemented in a software executable object. Alternatively, the processes, methods, or algorithms can be embodied in whole or in part using suitable hardware components, such as Application Specific Integrated Circuits (ASICs), Field-Programmable Gate Arrays (FPGAs), state machines, controllers or other hardware components or devices, or a combination of hardware, software and firmware components.

[0075] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to strength, durability, life cycle, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, to the extent any embodiments are described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics, these embodiments are not outside the scope of the disclosure and can be desirable for particular applications.

[0076] With regard to the processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating certain embodiments and should in no way be construed so as to limit the claims.

[0077] Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

[0078] All terms used in the claims are intended to be given their broadest reasonable constructions and their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary in made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.

[0079] The abstract of the disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

[0080] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

Claims

1. A method for providing audio to positions around a vehicle, comprising: storing a position mapping defining information indicative of how each of a plurality of closures of a vehicle, in a plurality of positions, affects sound emanating from a plurality of speakers of the vehicle at different positions around the exterior of the vehicle; tracking respective locations of at least one user outside the vehicle; and utilizing the position mapping to apply power levels to the plurality of speakers and positioning to the plurality of the closures to direct sound output from the speakers to the respective locations of the at least one user at a desired volume level.

2. The method of claim 1, wherein the plurality of positions includes a plurality of intermediate positions between an open position and a closed position.

3. The method of claim 1, wherein the plurality of closures includes one or more of a door, a window, a sunroof, a decklid, a liftgate or a tailgate.

4. The method of claim 1, further comprising: capturing, using one or more microphones, measurements of sound output received at a plurality of positions around the vehicle from the plurality of speakers of the vehicle over a plurality of different positions of the plurality of closures; and constructing the position mapping based on the measurements, such that the power levels to the plurality of speakers and the positioning of the plurality of the closures is indexed according to the plurality of positions and the measurements of the sound output.

5. The method of claim 4, further comprising: training a machine learning model, based on combinations of speaker power level, closure position, and the measurements of the sound output as ground truth; and inferring the power levels to the plurality of speakers and the positioning of the plurality of the closures using the machine learning model based on the respective locations of the at least one user and the desired volume level.

6. The method of claim 1, further comprising utilizing exterior lighting of the vehicle to operate as level meters, such that lighting intensity and/or color provided by the exterior lighting corresponds to relative loudness of the audio being played back.

7. The method of claim 1, wherein the at least one user includes a first user and a second user, and further comprising using at least a subset of the plurality of speakers to provide white noise to a respective location of the second user to give privacy to the first user.

8. The method of claim 1, further comprising: detecting a gesture using sensors of the vehicle; and adjusting the desired volume level based on the gesture.

9. The method of claim 1, further comprising repositioning the vehicle in addition to adjusting the positioning to the plurality of the closures to direct the sound output from the speakers to the respective locations of the at least one user at the desired volume level.

10. A vehicle for use in providing audio to positions around the vehicle, comprising: one or more controllers, configured to store a position mapping defining information indicative of how each of a plurality of closures of the vehicle, in a plurality of positions, affects sound emanating from a plurality of speakers of the vehicle at different positions around the exterior of the vehicle, the plurality of positions including a plurality of intermediate positions between an open position and a closed position, track respective locations of at least one user outside the vehicle, and utilize the position mapping to apply power levels to the plurality of speakers and positioning to the plurality of the closures to direct sound output from the speakers to the respective locations of the at least one user at a desired volume level.

11. The vehicle of claim 10, wherein the plurality of closures includes one or more of a door, a window, a sunroof, a decklid, a liftgate, or a tailgate.

12. The vehicle of claim 10, wherein the one or more controllers are further configured to: capture, using one or more microphones, measurements of sound output received at a plurality of positions around the vehicle from the plurality of speakers of the vehicle over a plurality of different positions of the plurality of closures; and construct the position mapping based on the measurements, such that the power levels to the plurality of speakers and the positioning of the plurality of the closures is indexed according to the plurality of positions and the measurements of the sound output.

13. The vehicle of claim 12, wherein the one or more controllers are further configured to: train a machine learning model, based on combinations of speaker power level, closure position, and the measurements of the sound output as ground truth; and infer the power levels to the plurality of speakers and the positioning of the plurality of the closures using the machine learning model based on the respective locations of the at least one user and the desired volume level.

14. The vehicle of claim 10, wherein the one or more controllers are further configured to utilize exterior lighting of the vehicle to operate as level meters, such that lighting intensity and/or color provided by the exterior lighting corresponds to relative loudness of the audio being played back.

15. The vehicle of claim 10, wherein the at least one user includes a first user and a second user, and the one or more controllers are further configured to use at least a subset of the plurality of speakers to provide white noise to a respective location of the second user to give privacy to the first user.

16. The vehicle of claim 10, wherein the one or more controllers are further configured to: detect a gesture using sensors of the vehicle; and adjust the desired volume level based on the gesture.

17. A non-transitory computer-readable medium comprising instructions for providing audio to positions around a vehicle that, when executed by one or more controllers of the vehicle, cause the vehicle to perform operations including to: store a position mapping defining information indicative of how each of a plurality of closures of the vehicle, in a plurality of positions, affects sound emanating from a plurality of speakers of the vehicle at different positions around the exterior of the vehicle, the plurality of positions including a plurality of intermediate positions between an open position and a closed position, track respective locations of at least one user outside the vehicle, and utilize the position mapping to apply power levels to the plurality of speakers and positioning to the plurality of the closures to direct sound output from the speakers to the respective locations of the at least one user at a desired volume level.

18. The medium of claim 17, wherein the one or more controllers are further configured to: capture, using one or more microphones, measurements of sound output received at a plurality of positions around the vehicle from the plurality of speakers of the vehicle over a plurality of different positions of the plurality of closures; and construct the position mapping based on the measurements, such that the power levels to the plurality of speakers and the positioning of the plurality of the closures is indexed according to the plurality of positions and the measurements of the sound output.

19. The medium of claim 18, wherein the one or more controllers are further configured to: train a machine learning model, based on combinations of speaker power level, closure position, and the measurements of the sound output as ground truth; and infer the power levels to the plurality of speakers and the positioning of the plurality of the closures using the machine learning model based on the respective locations of the at least one user and the desired volume level.

20. The medium of claim 17, wherein the one or more controllers are further configured to one or more of: utilize exterior lighting of the vehicle to operate as level meters, such that lighting intensity and/or color provided by the exterior lighting corresponds to relative loudness of the audio being played back; use, when the at least one user includes a first user and a second user, at least a subset of the plurality of speakers to provide white noise to a respective location of the second user to give privacy to the first user; and/or adjust the desired volume level based on a gesture captured using sensors of the vehicle.